

Instructor: Dr. Jim D. Pham

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Office Hrs.: By Appointment

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Section: CS-124-04 (094890; MW; 6:30PM; Fremont Campus, Rm 3 3405)

Course Description:

This course involves the study and implementation of data structure programming techniques. The emphasis is on the data structures of stacks, queues, lists, trees, and hash tables; the use of recursion; and the application of these tools primarily to searching and sorting. Students implement these concepts by writing programs in an object-oriented language such as C++.

Prerequisites:

Advisory: CS 102 Introduction to Computer Programming Using C++

Textbook:

Cay Horstmann Big C++ 3rd eText Edition, John Wiley & Sons Publishing, 2019

ISBN: 979-8-203-10904-0

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: OHLONECS124PhamSpring2025
3. Subscribe

Student Learning Outcomes

Students will be able to:

1. Apply a systematic approach to the design, construction, and management of computer programs, emphasizing programming style, documentation, and debugging techniques.
2. Demonstrate knowledge of the basic data structures of stacks, lists, trees, graphs, queues, and hash tables.
3. Implement (program) these structures in their appropriate applications, such as sorting and searching.
4. Compare and contrast the benefits and costs of different data structures in order to choose the best one for a particular application.

Course Content:

Apply a systematic approach to the design, construction, and management of computer programs, emphasizing programming style, documentation, and debugging techniques.

1. Review of Object-Oriented programming techniques
2. Discuss the significance of data abstractions and implementations.
3. Measuring the performance of algorithms.
4. The effects of scale on programming methodology
5. Lab: Program to introduce data structures and design techniques
6. Lab: Program to analyze algorithms A.

Demonstrate knowledge of the basic data structures of stacks, lists, trees, queues, and hash tables.

1. Review of linked lists and arrays

2. Trees as a special form of a linked list.
3. Adaptors built on lists and arrays: stacks, and queues.
4. Libraries of data structures: The C++ Standard Library
5. Lab: Program to explore basic structures including lists, stacks, and queues.

Implement (program) these structures in their appropriate applications, such as sorting and searching.

1. Implementation strategies for different data structures
2. Sequential and Binary search algorithms.
3. Quadratic sorting algorithms (Bubble, Insertion, and Selection)
4. $O(N \log N)$ sorting algorithms (Heapsort, Mergesort, and Quicksort)
5. Lab: Program to implement structures and compare the performance of several search/sort algorithms.

Compare and contrast the cost and benefits of different data structures.

1. Strategies for choosing the right data structure.
2. Big-O notation to classify algorithm performance.
3. Identifying best, average, and worst-case behaviors.
4. The concept of recursion.
5. Binary search trees and recursive procedures
6. Lab: Mathematical techniques to classify the performance of several algorithms

Methods of Evaluation:

1. The instructor will lead class discussions where students are called on to describe concepts. This is a formative assessment that is not graded but used to discover the points that need further clarification and exercises.
2. Required discussion posts from students and/or teams of students to evaluate reading comprehension. These are graded for completion and are also formative.
3. Regular quizzes require students to demonstrate current mastery of concepts. These are graded and summative assessments.
4. Lab exercises that implement (program) data structures in their appropriate applications, such as sorting and searching. These are summative and graded on the quality of work and including successful program results, testing, and documentation.
5. The final exam will assess their comprehensive mastery of the techniques covered in class.

Methods of Instruction:

1. Lectures: The presentation slides and assignments will be accessed through Ohlone Canvas. See Instructional Resources below.
2. Assignments: End-of-chapter activities and online activities will be assigned weekly to reinforce the material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Participation: Student participation will be graded by the level of class participation on the Discussion Board. Each student will be required to do a minimum of 2 posts (1 post and 1 reply) per chapter on the Canvas Discussion Board.

Instructional Resources:

Canvas: Class documents, syllabus, lectures, assignments, and discussions will be posted on Canvas. You access Canvas at this URL: <https://ohlone.instructure.com>.

Email:

If you send me an email, then please put the **course number – name – topic** into the email **Subject** field. For example, CS 124 – Name – Issue

Discussion Board:

Your participation on the Discussion Board (i.e., Canvas) will be an important aspect of the class. Students will be required to post and respond to at least two posts on the discussion board.

Labs and Assignments:

Lab programming assignments will be assigned each week. It is the student's responsibility to submit the class assignments at the due date.

Policy on Student Work:

Labs and assignments are to be done individually unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

GRADING:**GRADING SCALE:**

Class Participation:	10%	A = 90-100
Quizzes:	10%	B = 80-89
Labs:	40%	C = 65-79
Midterm:	20%	D = 55-64
Final Exam:	20%	F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	01/27/2025 01/29/2025	zyBooks Chapters 1 and 2: Classes, Inheritance, and Composition - Lab 1
2	02/03/2025 02/05/2025	zyBooks Chapter 3: Advanced C++, Overloading, Operators, etc. - Lab 1 continuing
3	02/10/2025 02/12/2025	zyBooks Chapter 4: Introduction to Data Structures and Algorithms - Lab 2
4	02/17/2025 02/19/2025	Presidents' Weekend (02/14 – 02/17) zyBooks Chapter 5: Array and Vector, and ADT - Lab 2 continuing
5	02/24/2025 02/26/2025	zyBooks Chapter 6: Lists - Lab
6	03/03/2025 03/05/2025	zyBooks Chapters 8: Linked Lists - Lab 3
7	03/10/2025 03/12/2025	zyBooks Chapters 9/10: Stacks and Queues - Lab 3 continuing
8	03/17/2025 03/19/2025	zyBooks Chapters 11-13: Sorting and Searching; and Recursion - Lab 3 Midterm (Chapter 1 – Chapter 9)
9	03/24/2025 03/30/2025	Spring Break (03/24 – 03/30)
10	03/31/2025 04/02/2025	zyBooks Chapters 17-21: Sets, Maps, and Hash Tables - Lab 4
11	04/07/2025 04/09/2025	zyBooks Chapters 17-21: Sets, Maps, and Hash Tables (continuing) - Lab 4 continuing
12	04/14/2025 04/16/2025	zyBooks Chapter 22: Tree Structures - Lab 5
13	04/21/2025 04/23/2025	zyBooks Chapter 25: B-Trees - Lab 5 continuing
14	04/28/2025 04/30/2025	zyBooks Chapter 25: B-Trees continuing - Lab 5 continuing
15	05/05/2025 05/07/2025	zyBooks Chapter 34: Graph - Special Lab assignment
16	05/12/2025 05/14/2025	zyBooks Chapter 34: Graph continuing - Special Lab assignment
17	05/19/2025	Final Exam